Wireshark

Monitor your network with greater ease...

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Wireshark is a free and open source network packet analysing software, which was known as Ethereal till 2006. If you're a developer, you can monitor network connections made by your applications. If you have a network or if your Internet connection is constantly getting congested, then you can use Wireshark to examine the cause. There are chances that you might find a virus or worm to be cause of the congestion. It also makes a very good educational tool for networking and security enthusiasts.

Selecting The Interface And Capturing

Wireshark can be used to capture data from all kinds of network adapters. If you have more than one network adapter, you need to first select it. Go to `Capture > Options`. Choose the one you want to capture from the dropdown menu and make changes to the settings if you need to. Click on `Start` to capture data. After you finish capturing data, click on the `Stop` button on the toolbar, or from the Capture menu.

Wireshark can also load data captured using other network analysing tools. Go to `File > Open` and select the file to be imported. You can also set filters during the import process. Captured data can be saved using `File > Save As`.

Understanding The Interface

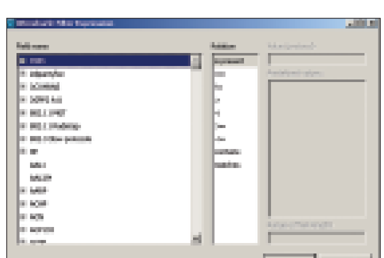
Working with Wireshark can be a very harrowing experience when you first use it. The three most important panes that you see from top to bottom are the packet info, packet details and packet bytes panes. The topmost pane displays the different packets passing through the network and the packet details shows important statistics and information on the selected packet. The last packet bytes pane displays the data in the packet itself. Clicking on a pane in the packet info pane shows the necessary information for that packet in the other two panes.

If you do not like the layout of the interface, you can change it by selecting `Edit > Preferences`. Click on `Layout` under the `User Interface` section on the left. Now click on one of the layouts and click `OK` to enable that layout.

Filtering In Wireshark

There is way too much data that Wireshark churns up when you set it to capture data. It is close to impossible to use all of that data. The filter feature is one of the most important things to understand in Wireshark, as it will make going through connections and packets much simpler than if you were to do it one packet at a time.

The simplest way to filter data is to use the Filter toolbar. Use keywords like tcp or udp to filter out data. More advanced parameters can be found by clicking on `Expression`. Scroll down to the category you want to filter. Expand the item and choose the parameter. Select a relation and enter a value. Click `OK` and the filter will be entered. A history of filters is stored in the same dropdown menu, so you can access the most used filters from time to time.



Filtering data

Colouring rules

To make tracking of packets and connections easier, Wireshark displays important events in unique colours. These colours can be changed and new ones added by clicking on `View > Coloring Rules`. You can create your own custom colouring rule by clicking on the `New` button. Give a name to the rule and enter a filter. Choose a background and foreground colour by clicking on the buttons below. Click `OK`. These rules can be saved and loaded back again using the Export and Import buttons.

For colouring of certain connections, select `View > Colorize Conversation`. You can also use the shortcuts `[Ctrl] + [1]` to `[0]` to set colours. The sequence of packets and transactions will be highlighted in that colour making it easy to find and understand.

Refresh Rate

Depending on the size of the network and the amount of data passing through it, Wireshark can significantly slow down your computer. All the packets are being updated on the interface in real time. Click on `Edit > Preferences > Capture`. Uncheck the box for and updated list of the packets in real time. Click OK. All the packets will now be captured in the background. When the command for stopping the data capture is given, you can see all captured data on the screen.

You can see the captured data in real-time by scrolling it on the window. If you find this annoying, you can disable it by unchecking the `Auto Scroll in Live Capture` item under `View`. ■

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